Reg. No										

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AMTCA101 – MATHEMATICAL FOUNDATIONS

Time : 3 hrs **Max Marks :75**

SECTION – A (5X2=10)

Answer **ALL** Questions

1. Define biconditional.
2. Define set.
3. Write down the of any two binary operations.
4. Define orthogonal matrix. Give an example.
5. State-cayley-hamilton theorem.

SECTION – B (3X5=15)

Answer any **THREE** Questions

6. Find the truth table of $(p \wedge 7q) \vee r$.
7. Let f&g be functions defined by $f(x) = 3x + 4$ and $g(x) = x^2 + 2$, find the formulate determining i) gof & ii) fog
8. In how many different ways can the letters of the word "POSSESSIVE" be arranged? In how many of these will the S's come together.

9. Find the Rank of matrix.

$$A=\begin{bmatrix} 6 & 1 & 3 & 8 \\ 4 & 2 & 6 & -1 \\ 10 & 3 & 9 & 7 \\ 16 & 4 & 12 & 15 \end{bmatrix}.$$

10. Find the matrix of transformation that maps points into points by reflection about the line y=*x*.

SECTION – C (5X10=50)

Answer **ALL** Questions

11. a) Prove that $(p \wedge q) \rightarrow r \equiv (p \rightarrow r) \vee (q \rightarrow r)$.

(or)

b) Test the validity of the argument, $p \rightarrow 7q, r \rightarrow p, \vdash 7r$.

12. a) Verify De – Morgan's laws using venn diagram.

(or)

b) If $f(x) = x^2 + x + 1$, g(*x*)=$2x-1$ where *x* is a real number. Find fog, gof, fof & gog.

13. a) Let S be a set and * be a binary operation on S satisfying the conditions

$a * a = a \ \forall a \in s$ and $(a * b) * c = (b * c) * a \quad \forall\ a, b, c \in$ S

Show that * is both commutative and associative.

(or)

b) Prove that i) $\dfrac{ncr}{nc_{r-1}} = \dfrac{n-r+1}{r}$ ii) $ncr + nc_{r-1} = (n+1)cr$.

14. a) Define unitary matrix. Show that $\begin{bmatrix} \frac{1+i}{2} & \frac{-1+i}{2} \\ \frac{1+i}{2} & \frac{1-i}{2} \end{bmatrix}$ is unitary.

(or)

b) If the matrix A is given by, $A = \begin{bmatrix} 1 & 1 & 1 \\ 2 & -1 & 3 \\ 3 & 2 & -1 \end{bmatrix}$ obtain a matrix B such that, AB=BA=1.

15. a) Find the matrix of transformation that maps into
i) Reflection about *x*-axis
ii) Reflection about y-axis
iii) Translation

(or)

b) Find the characteristic equation of $A = \begin{bmatrix} 1 & 1 & 3 \\ 5 & 2 & 6 \\ -2 & -1 & -3 \end{bmatrix}$ & show that the matrix A satisfies the equation.

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